

Deep Learning for Long-Stride Protein Molecular Dynamics Simulation

Representation, Dynamics Model Development, and Anti-Noise Validation Plan

Prepared as a research-program blueprint for developing a transferable protein dynamics world model.

Central objective	Learn a transferable stochastic transition operator that advances protein conformational ensembles over long MD strides.
Core idea	Use inference as a long stride over molecular time while preserving structural validity, equilibrium statistics, and kinetic observables.
Key technical pillars	Hierarchical protein representation; stochastic long-stride trajectory inference; physics- and kinetics-aware training.
Primary risk	The model may learn local thermal noise or visually plausible structures instead of meaningful slow dynamics.
Primary safeguard	Separate slow and fast latent variables; train on multi-lag data; enforce VAMP/Koopman, Chapman-Kolmogorov, equilibrium, and null-control tests.

One-line model statement

protein graph + residue frames + torsions -> slow latent dynamics -> stochastic long-stride transition -> geometry-constrained all-atom decoder -> physics correction

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1. Executive summary

The proposed research direction is to develop a deep learning system that enables long-stride molecular dynamics simulation for proteins. Conventional MD integrates equations of motion with femtosecond-scale timesteps. Many biologically important processes - allosteric transitions, domain opening, folding, ligand binding, and functional loop rearrangements - happen over nanosecond to millisecond timescales or beyond. The central idea is to train a deep model to learn implicit protein dynamics so that inference can jump forward by an integer stride of many MD steps.

The appropriate object to learn is not a deterministic mapping from the current frame to a single future frame. At long stride, many futures are valid. The model should learn a conditional stochastic transition kernel:

$$p_{\theta}(x_{t+\tau} | x_t, h_t, \tau, c)$$

Here x_t is the protein state, h_t is optional history or latent memory, τ is the long stride, and c contains physical and biological conditions such as sequence, topology, temperature, force field, solvent/ion context, ligand state, mutation state, and pH/protonation proxies.

The model should be developed as a protein dynamics world model: a model that learns transferable physical principles across proteins rather than needing to learn each protein independently. The strongest scientific claim would be that the model learns a physically constrained transition operator over conformational ensembles, preserving thermodynamic and kinetic structure rather than simply generating plausible structures.

The plan has three major components: (1) a multi-resolution protein representation based on residue frames, torsions, dynamic interaction graphs, and slow latent variables; (2) a stochastic long-stride dynamics model using flow matching, diffusion, latent SDEs, or transfer-operator modules; and (3) a strict validation framework proving that the model learns meaningful dynamics instead of noise.

2. Research hypothesis and target claim

Central hypothesis:

A transferable protein dynamics model can learn a physically constrained stochastic transition operator over protein conformations, enabling inference-time jumps over many MD integration steps while preserving equilibrium structures, metastable transitions, and kinetic observables.

This hypothesis decomposes into three experimentally testable subclaims.

Subclaim	Meaning	How to test it
Representation claim	Protein conformations should be represented using symmetry-respecting, chemically constrained, multi-resolution variables rather than raw Cartesian coordinates alone.	Compare Cartesian, torsion, residue-frame, contact-map, and hybrid representations using reconstruction quality, VAMP score, kinetic observables, and transfer performance.
Dynamics claim	Long-stride inference should be stochastic and ensemble-based, not deterministic coordinate regression.	Evaluate conditional ensembles, free-energy surfaces, transition matrices, implied timescales, and multi-step rollout stability.
World-model claim	Training across many proteins and conditions can teach reusable physical regularities.	Split test sets by trajectory, protein, sequence family, fold, temperature, length, mutation, ligand state, and force field.

3. Background and reference points

Several recent studies support the feasibility of this direction, but also show that transferable, physically calibrated, long-horizon protein dynamics remains an open challenge. Timewarp used a normalizing flow

as a proposal distribution in MCMC and learned time-coarsened moves of about 10^5 to 10^6 fs for small peptides. mdCATH provides a large all-atom MD dataset across thousands of protein domains with multiple temperatures, coordinates, and forces. DeepJump applies Euclidean-equivariant flow matching to protein conformational dynamics across multiple temporal scales. STAR-MD frames long-horizon protein dynamics as scalable spatiotemporal SE(3)-equivariant diffusion. VAMPnets and GraphVAMPNet motivate learning slow kinetic coordinates rather than relying on hand-crafted Markov-state-model features. EGNN, SE(3)-equivariant attention, NequIP-like ideas, AlphaFlow, and MDGen motivate geometric and generative architectures for molecular systems.

Reference point	Relevant lesson for this project
Timewarp	Use deep learning for large MD proposals, but consider statistical-mechanical correction rather than unconstrained generation.
mdCATH	World-model training needs diverse proteins, temperatures, coordinates, forces, and carefully separated train/test splits.
DeepJump	Equivariant flow matching is a natural candidate for multi-scale protein conformational transitions.
STAR-MD	Long-horizon rollout requires joint spatiotemporal modeling and control of error accumulation.
VAMPnets / GraphVAMPNet	Meaningful dynamics should be evaluated through slow kinetic variables, transition matrices, and implied timescales.
EGNN / SE(3)-equivariant models	Architectures should respect rotations, translations, and permutation symmetries.
AlphaFlow / ESMFlow	Flow matching can generate protein ensembles and can be trained on MD-derived conformational flexibility.
MDGen	Generative trajectory models can support forward simulation, transition path sampling, and trajectory upsampling.

4. Representation development plan

The representation is the foundation of the project. A poor representation will make the model learn rotation, translation, bond geometry, and local noise rather than transferable dynamics. The recommended strategy is a hierarchical representation that combines backbone residue frames, side-chain torsions, dynamic interaction graphs, and learned slow variables.

4.1 Representation principles

- Respect physical symmetries: translation, rotation, and permutation of equivalent atoms should not change the physical state.
- Use internal coordinates where possible: torsions, distances, local frames, contacts, and pairwise orientations are more meaningful than raw coordinates alone.
- Separate slow and fast information: slow variables encode metastable states and functional motion; fast variables encode thermal vibration and local roughness.
- Preserve chemical validity: topology, bond lengths, bond angles, peptide planarity, chirality, and side-chain rotamers should be built into the representation or decoder.
- Condition on the environment: temperature, force field, solvent model, ions, ligand state, mutation state, and stride length should be explicit conditioning variables.

4.2 Full protein state

For each frame, define the protein state as:

$$\mathbf{x}_t = \{A, R_t, V_t, F_t, B, G_t, S, C\}$$

Symbol	Meaning
A	Atom identity table: element, atom name, residue name, residue index, chain ID, mass, charge proxy, backbone/side-chain flag.
R_t	Coordinates at time t. Early versions may use backbone or heavy atoms; later versions can include all atoms.
V_t	Velocities, when available. Useful for short memory, but long-stride velocities decorrelate quickly.
F_t	Forces, when available. Useful for auxiliary force/energy learning and local physical principles.
B	Static covalent graph: bonds, angles, improper torsions, peptide connectivity, residue topology.
G_t	Dynamic interaction graph: contacts, hydrogen bonds, salt bridges, spatial neighbors, ligand/protein edges.
S	Sequence and residue-level annotations: residue type, secondary structure, solvent exposure, domain assignment, disorder/flexibility prior.
C	Global conditions: temperature, force field, solvent model, ion concentration, pH/protonation proxy, ligand state, mutation state, stride tau.

4.3 Layer 1: residue-frame backbone representation

For each residue i , define a local coordinate frame from backbone atoms, for example N, CA, and C. Represent the backbone as local SE(3) residue-frame transformations plus internal torsions:

$$B_t = \{T_i(t), \phi_i(t), \psi_i(t), \omega_i(t)\} \text{ for residues } i = 1, \dots, N$$

This representation captures global domain motion and local backbone flexibility while reducing the burden of learning rigid-body invariance. It should be the primary representation for long-stride backbone dynamics.

Backbone feature	Purpose
Residue type	Amino-acid identity and local chemical context.
Sequence position / relative position	Local sequence context and chain order.
N-CA-C-O frame	Local backbone orientation and residue coordinate system.
phi, psi, omega torsions	Internal backbone degrees of freedom.
Secondary structure	Helix/sheet/coil context and structural stability.
Solvent exposure	Proxy for packing, hydration, and surface flexibility.
Local strain	Deviation from ideal geometry or high-energy local conformation.
Disorder/flexibility prior	Useful for loops, termini, and intrinsically disordered regions.

4.4 Layer 2: side-chain torsion representation

Represent side chains through chi torsions rather than raw side-chain atom positions:

$$S_t = \{\chi_{i,1}(t), \chi_{i,2}(t), \chi_{i,3}(t), \chi_{i,4}(t)\}$$

This is more chemically meaningful because side-chain motion is dominated by rotameric torsions. The development should follow a coarse-to-fine path: backbone transition -> side-chain torsion transition -> all-atom reconstruction.

Side-chain feature	Purpose
Residue type	Determines number and chemistry of chi angles.
Current chi angles	Current rotameric state.
Local backbone frame	Conditions the side-chain coordinate system.
Neighbor residue identities	Packing and steric context.
Contact neighborhood	Local interactions and buried/surface state.
Hydrogen-bond/salt-bridge state	Functional side-chain constraints.

4.5 Layer 3: dynamic interaction graph

The graph should contain both static and dynamic edges:

$$G_t = (V, E_{\text{bond}}, E_{\text{seq}}, E_{\text{spatial}}(t), E_{\text{contact}}(t), E_{\text{HB}}(t), E_{\text{ionic}}(t), E_{\text{ligand}}(t))$$

Edge type	Static or dynamic?	Purpose
Covalent bond edges	Static	Preserve chemistry and topology.
Sequence-neighbor edges	Static	Encode chain continuity and local backbone context.
k-nearest spatial edges	Dynamic	Represent current packing and local environment.
Contact edges	Dynamic	Represent metastable states and rearrangements.
Hydrogen-bond candidate edges	Dynamic	Track secondary-structure and specific interactions.
Salt-bridge edges	Dynamic	Track charged residue coupling and electrostatic switching.
Ligand/protein edges	Dynamic	Support holo/apo and induced-fit extensions.
Domain/domain edges	Dynamic	Support allostery and large-scale motions.

Use soft contact probabilities in addition to hard distances:

$$C_{ij}(t) = \text{sigmoid}((r_c - D_{ij}(t)) / \alpha)$$

Soft contacts are useful because they change smoothly and capture slow metastable rearrangements better than raw coordinates alone.

4.6 Layer 4: learned slow variables

For long-stride dynamics, the most important state variables are often slow collective variables rather than high-frequency atomistic vibrations. Learn:

$$z_t = E_{\theta}(x_t), \text{ with } z_t = [z_t^{\text{slow}}, z_t^{\text{fast}}]$$

Slow variable candidate	Why it matters
Domain opening angle	Captures hinge and allosteric motions.
Inter-residue contact map	Encodes packing states and metastability.
Secondary-structure state	Tracks helix/sheet stability and unfolding.
Pocket volume	Connects dynamics to binding and function.
Salt-bridge state	Captures electrostatic switches.
Hydrogen-bond network	Captures backbone and side-chain stabilization.
RMSD to metastable states	Provides interpretable state labels.
Learned VAMP/Koopman coordinates	Optimized for slow kinetic prediction.

5. State schema and data objects

A robust implementation should store each MD frame as standardized tables. This allows representation ablation, reproducible training, and dataset integration.

5.1 Atom table

Field	Example or meaning
atom_index	0, 1, 2, ...
atom_name	N, CA, C, O, CB, CG, ...
element	C, N, O, S, H, ...
residue_index / residue_name	Residue position and amino-acid identity.
chain_id	Protein chain identifier.
mass / charge proxy	Physical features for dynamics and force prediction.
is_backbone	Boolean flag for backbone vs side-chain atom.
coordinates	3D vector at time t.
velocity	Optional 3D vector.
force	Optional 3D vector.

5.2 Residue table

Field	Example or meaning
residue_index / residue_type	Residue order and amino-acid identity.
local frame	Rotation R_i and translation t_i .
phi, psi, omega	Backbone torsion angles.
chi_1, chi_2, ...	Side-chain torsion angles where present.
secondary structure	Helix, sheet, coil, turn, or probability vector.
solvent exposure	SASA or learned/binned exposure feature.
local flexibility	RMSF, B-factor proxy, or learned flexibility prior.
domain ID	Optional domain or segment assignment.

5.3 Edge table

Field	Example or meaning
sender, receiver	Residue or atom pair.
edge type	bond, sequence, contact, hydrogen bond, salt bridge, ligand contact.
distance	Pairwise distance or transformed distance feature.
relative orientation	Frame-to-frame transform for residue pairs.
contact probability	Soft contact indicator.
lifetime feature	Recent persistence of the interaction.
energy proxy	Optional approximate interaction score.

5.4 Global condition table

Field	Purpose
temperature	Controls fluctuation amplitude, unfolding tendency, and rates.
force field	Needed to handle simulation protocol bias.
solvent model	Explicit, implicit, or coarse descriptors.
ion concentration	Affects charged interactions and screening.
pH/protonation proxy	Affects histidine, acidic/basic residues, and salt bridges.
stride tau	Defines the prediction/inference horizon.
system class	Apo, holo, mutant, complex, folded, unfolded, biased.
protein family/fold	Useful for analysis and split control; should not cause leakage.

5.5 Representation development stages

Stage	Representation	Purpose
R0	Preprocessing and normalization	Remove PBC artifacts, standardize atom naming, compute torsions, frames, contacts, secondary structure, hydrogen bonds, and split data by protein/fold.
R1	Backbone-only	Use N, CA, C, O, residue frames, phi/psi/omega, residue graph, temperature, and stride; first prototype.
R2	Side chains	Add chi angles and side-chain heavy-atom reconstruction.
R3	Forces and velocities	Add forces and velocities as auxiliary physical features. Treat velocity as short-memory information.
R4	Solvent and ligand context	Add solvent exposure, local water density, ion-neighbor features, ligand contacts, pocket volume.

6. Dynamics model development plan

The dynamics model should learn a stochastic long-stride transition kernel rather than a deterministic coordinate predictor. At long lag time, exact atomic prediction is impossible and physically undesirable. The correct target is a conditional distribution of plausible future conformations.

$$x_{t+\tau} \sim p_{\theta}(x_{t+\tau} | x_t, h_t, \tau, c)$$

6.1 Overall architecture

$x_t \rightarrow$ multi-resolution encoder $\rightarrow z_{t_slow}, z_{t_fast} \rightarrow$ stochastic propagator $\rightarrow z_{t+\tau_slow}, z_{t+\tau_fast} \rightarrow$ geometry-constrained decoder $\rightarrow \hat{x}_{t+\tau}$

Module	Role
Representation encoder	Converts coordinates, topology, sequence, forces, velocities, and conditions into equivariant/geometric features.
Slow latent dynamics module	Learns metastable transitions, contact rearrangements, domain motion, folding/unfolding coordinates, and allostery.
Fast stochastic module	Models local thermal fluctuations, side-chain breathing, loop wiggles, and unresolved solvent effects.
Geometry/physics decoder	Generates valid backbone, side-chain, and all-atom structures with chemical constraints.
Correction module	Filters, relaxes, or reweights proposed states to improve thermodynamic and structural validity.

6.2 Encoder

The encoder maps the current state into slow and fast latent variables:

$$E_{\theta}(x_t) \rightarrow (z_{t_slow}, z_{t_fast}, q_t)$$

Latent	Meaning
z_{t_slow}	Metastable state, contact pattern, domain orientation, folding coordinate, allosteric state.
z_{t_fast}	Local torsion fluctuation, side-chain noise, thermal vibration, loop roughness.
q_t	Uncertainty, confidence, and out-of-distribution signal.

The encoder should combine local atomistic message passing, residue-frame equivariant attention, contact-graph message passing, global attention, and condition embeddings. Global attention is especially important for domain-domain and allosteric coupling because local graph layers alone can oversquash long-range information.

6.3 Slow latent propagator

The slow propagator should model metastable evolution:

$$z_{t+\tau_slow} \sim K_{\theta_slow}(z_{t+\tau_slow} | z_{t_slow}, \tau, c)$$

Candidate model	Use case
Flow matching	Efficient conditional generation of future conformational states.
Diffusion model	Stable multimodal generation with robust uncertainty.
Conditional normalizing flow	Likelihood, reweighting, and MCMC proposal generation.
Latent SDE	Natural stochastic dynamics model for thermally driven systems.
Koopman/VAMP module	Kinetic interpretability and slow timescale preservation.

Candidate model	Use case
Mixture transition network	Explicit metastable-state jumps and multimodal futures.

Recommended design: use flow matching or diffusion for generative transitions, plus a VAMP/Koopman auxiliary head for slow kinetic structure.

6.4 Fast stochastic module

The fast module should explain local decorrelated noise without dominating the representation:

$$z_{\{t+\tau\}}_{\text{fast}} \sim N(\mu_{\text{theta}}(z_{t_{\text{slow}}}, c, \tau), \Sigma_{\text{theta}}(z_{t_{\text{slow}}}, c, \tau))$$

The fast module captures backbone vibration, rotamer breathing, loop roughness, and contact flickering. It should be conditioned on the slow latent so that local fluctuations are compatible with the current metastable state.

6.5 Decoder

The decoder should generate structures coarse-to-fine:

$z_{\{t+\tau\}}_{\text{slow}}, z_{\{t+\tau\}}_{\text{fast}} \rightarrow$ backbone frames $\rightarrow$ backbone torsions $\rightarrow$ side-chain torsions $\rightarrow$ all-atom coordinates

Decoder stage	Output	Constraint
Backbone frame decoder	Residue SE(3) frames	Equivariance and chain continuity.
Backbone torsion decoder	phi, psi, omega	Ramachandran validity and peptide planarity.
Side-chain decoder	chi angles	Rotamer validity and packing.
All-atom builder	Heavy atoms and optional hydrogens	Bond lengths, bond angles, chirality.
Relaxation head	Corrected coordinates	Reduced clashes and local strain.

6.6 History and non-Markovian effects

All-atom MD is Markovian in positions and velocities, but protein-only dynamics after removing explicit solvent are generally non-Markovian. Therefore, use either positions plus velocities, a short history window, or a recurrent/transformer memory. Recommended initial choice:

$$h_t = \text{Transformer}(x_{\{t-3\Delta\}}, x_{\{t-2\Delta\}}, x_{\{t-\Delta\}}, x_t)$$

A short history helps infer whether a domain is opening, closing, unfolding, or simply fluctuating. However, the model should avoid relying on long history as a memorization pathway.

7. Inference strategy

7.1 Ensemble inference

Long-stride inference should be ensemble-based. For one input structure, generate multiple futures:

$$\{x_{\text{hat}}_{t+\tau}^{\{(1)\}}, x_{\text{hat}}_{t+\tau}^{\{(2)\}}, \dots, x_{\text{hat}}_{t+\tau}^{\{(K)\}}\}$$

A single long-stride future is not meaningful because the dynamics are stochastic and many conformational futures can be physically valid. The model should output a conditional ensemble, uncertainty, and optionally an estimated transition probability or weight.

7.2 Practical inference loop

- 1 Encode x_t using residue-frame, torsion, graph, and condition features.
- 2 Sample K slow latent futures from the slow transition model.
- 3 Sample fast fluctuations conditional on each slow future.
- 4 Decode coarse backbone frames and backbone torsions.
- 5 Decode side-chain torsions and rotamers.
- 6 Reconstruct all-atom coordinates.
- 7 Apply geometry filters for bonds, angles, chirality, peptide planarity, Ramachandran validity, and clashes.
- 8 Apply energy, force, or learned validity filters.
- 9 Optionally run short restrained MD or energy minimization.
- 10 Keep accepted structures and branch each accepted state for the next long stride.

7.3 Predictor-corrector inference

For high-quality simulation acceleration, use the neural model as a proposal generator rather than an unconstrained simulator:

$x_t \rightarrow$ neural long-stride proposal $\rightarrow$ geometry check $\rightarrow$ energy check $\rightarrow$ short relaxation $\rightarrow$ accept/reweight

Correction mechanism	Purpose
Geometry projection	Fix broken bond geometry, chirality, peptide planarity, and chain continuity.
Energy minimization	Remove severe clashes and local strain.
Short restrained MD	Relax local structure while preserving the proposed slow move.
Learned reweighting	Correct generated ensemble bias using an approximate energy or density ratio.
Metropolis-Hastings	Enforce a target equilibrium distribution where a tractable proposal density and energy are available.

7.4 Endpoint generator versus path generator

Option	Pros	Cons	Recommendation
Endpoint generator: $x_t \rightarrow x_{t+\tau}$	Simpler, faster, good for enhanced sampling and proposals.	May miss path mechanism and detailed transition route.	Start here.

Option	Pros	Cons	Recommendation
Path generator: $x_t \rightarrow x_{t+\Delta}, \dots, x_{t+\tau}$	Better mechanistic interpretability and trajectory movies.	More expensive and more prone to rollout drift.	Add after endpoint generator works.

8. Training objectives

The model should use a composite loss. Coordinate MSE alone is insufficient and can be misleading at long stride because it averages over multiple valid futures.

$$L = \lambda_{\text{gen}} L_{\text{gen}} + \lambda_{\text{geom}} L_{\text{geom}} + \lambda_{\text{kin}} L_{\text{kin}} + \lambda_{\text{CK}} L_{\text{CK}} + \lambda_{\text{eq}} L_{\text{eq}} + \lambda_{\text{phys}} L_{\text{phys}} + \lambda_{\text{unc}} L_{\text{unc}}$$

8.1 Generative transition loss

Train the stochastic transition distribution rather than a single future coordinate:

$$L_{\text{gen}} = -\log p_{\theta}(x_{\{t+\tau\}} | x_t, \tau, c)$$

Depending on the model, this can be a negative log likelihood, flow-matching loss, diffusion score-matching loss, or variational objective.

8.2 Geometry loss

Use internal-coordinate and relational geometry losses rather than only aligned Cartesian RMSD:

$$L_{\text{geom}} = \lambda_d ||D - D_{\text{hat}}||^2 + \lambda_{\text{angle}} ||a - a_{\text{hat}}||^2 + \lambda_{\text{torsion}} ||\theta - \theta_{\text{hat}}||^2 + \lambda_{\text{frame}} ||T - T_{\text{hat}}||^2$$

Include bond length, bond angle, peptide plane, chirality, Ramachandran, rotamer, steric clash, and frame-consistency losses.

8.3 Kinetic loss

Apply a VAMP/Koopman-style objective to the slow latent variables. The goal is for $z_{t,\text{slow}}$ to retain time-predictive information at lag τ while ignoring fast decorrelated noise.

$$z_{t,\text{slow}} = E_{\theta}(x_t), z_{\{t+\tau\},\text{slow}} = E_{\theta}(x_{\{t+\tau\}})$$

This loss should be applied specifically to the slow latent space, not to the full coordinate representation.

8.4 Chapman-Kolmogorov consistency

The learned transition operator should compose consistently across strides:

$$K_{\{\tau_1 + \tau_2\}} \sim K_{\{\tau_2\}} K_{\{\tau_1\}}$$

This prevents the model from learning independent, stride-specific generators that do not behave like a dynamical system.

8.5 Equilibrium distribution loss

Generated ensembles should match MD equilibrium observables:

Observable	Purpose
RMSF	Residue flexibility.
Contact probability map	Packing and metastable state populations.
Pairwise distance distributions	Global ensemble structure.
Torsion distributions	Local conformational statistics.
Secondary-structure populations	Helix/sheet/coil stability.
Free-energy surfaces	Thermodynamic landscape along collective variables.
Metastable-state populations	Equilibrium weights of functional states.

8.6 Physical regularization

Add physical penalties and constraints:

$$L_{\text{phys}} = L_{\text{bond}} + L_{\text{angle}} + L_{\text{clash}} + L_{\text{chirality}} + L_{\text{energy}} + L_{\text{temperature}}$$

If forces are available, add a force auxiliary head. This encourages learning local physical principles, not just statistical correlations.

9. Ensuring meaningful dynamics instead of noise

This is the central technical challenge. Thermal noise is real, but useful long-stride dynamics live in slow, persistent, predictive variables. The model must learn both without confusing them. The safeguards should operate at five levels: data design, lag-time design, representation design, loss design, and validation/null controls.

9.1 Lag-time design

Do not train only on adjacent frames. Adjacent frames mostly contain high-frequency vibration. Train across a lag schedule such as:

```
tau in {10 ps, 100 ps, 1 ns, 10 ns, 100 ns, 1 microsecond}, depending on data availability
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Lag regime	Dominant signal
fs-ps	Bond vibration and local thermal noise.
10-100 ps	Side-chain breathing and loop motion.
1-10 ns	Contact flickering and local rearrangement.
10-100 ns	Rotamer transitions and loop-state transitions.
100 ns-microsecond	Domain motion and metastable transitions.
microsecond+	Folding, allostery, rare events, ligand binding/unbinding.

9.2 Choose lags using autocorrelation analysis

Before training, compute autocorrelation functions for candidate observables:

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$$C_f(\tau) = E[(f(x_t) - \text{mean}_f)(f(x_{t+\tau}) - \text{mean}_f)] / \text{Var}(f)$$

```

Use backbone torsions, side-chain torsions, contact indicators, inter-residue distances, secondary-structure states, pocket volume, domain angle, hydrogen-bond occupancy, and salt-bridge occupancy. Choose tau such that fast vibration has decayed but slow processes retain predictive information.

```
tau_train > tau_fast_noise, but tau_train < tau_slow_process_decorrelation
```

9.3 Data sampling against noise

Training pairs should be stratified over protein, temperature, RMSD, contact state, secondary-structure state, metastable cluster, and lag time. Avoid overrepresenting near-identical frames, otherwise the model will learn local fluctuations. Include rare-transition data through high-temperature simulations, replica exchange, umbrella sampling, metadynamics, transition path sampling, folding/unfolding trajectories, ligand unbinding trajectories, and active-learning-generated MD. Biased or high-temperature data must be explicitly labeled by condition.

9.4 Slow/fast latent separation

Explicitly split the latent state:

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$$z_t = [z_{t\_slow}, z_{t\_fast}]$$

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z_{slow} should encode	z_{fast} should encode
Contact map, domain orientation, pocket opening, secondary-structure state, hydrogen-bond network, salt-bridge state, metastable cluster, folding coordinate.	Small coordinate fluctuations, local side-chain noise, loop roughness, thermal vibration, and contact flickering.

The slow latent should be time-predictive and kinetically regularized. The fast latent should be stochastic, local, and allowed to decorrelate.

9.5 Auxiliary slow-observable prediction

Compute slow observables for each frame and train the slow latent to predict future observables:

$$\mathbf{o}_t = \{C_{ij}(t), D_{ij}(t), \phi_i(t), \psi_i(t), \chi_i(t), SS_i(t), \text{pocket_volume}(t), Q(t)\}$$

$$\mathbf{o}_{\hat{t}+\tau} = g_{\theta}(z_{t_slow}, \tau)$$

This forces z_{slow} to encode variables that matter for future dynamics rather than irrelevant instantaneous vibration.

9.6 Anti-noise losses

Loss or constraint	Purpose
Time-lagged contrastive loss	Identify true future states among hard negative samples.
VAMP/Koopman loss	Select features that preserve slow kinetic information.
Lagged mutual information	Maximize $I(z_{t_slow}; z_{t+\tau_slow})$ while limiting fast-noise information.
Multi-step rollout loss	Prevent one-step plausibility from degrading under repeated inference.
Noise-injection consistency	Prevent the model from overreacting to small irrelevant perturbations.
Semigroup/CK consistency	Force transition operators to compose like dynamics.
Detailed-balance regularization	Encourage reversible equilibrium dynamics in slow-state space.

For time-lagged contrastive learning, use the true future as the positive pair and hard negatives from the same protein, same fold, wrong metastable state, wrong temperature, or time-shuffled frames. Random negatives that are obviously different are too easy.

$$L_{\text{contrast}} = -\log \exp(\text{sim}(z_t, z_{t+\tau}))/\beta) / \sum_j \exp(\text{sim}(z_t, z_{j_negative}))/\beta)$$

10. Validation and anti-noise controls

10.1 Four levels of validation

Validation class	Metrics
Structural validity	Bond length distributions, bond angle distributions, chirality violation rate, clash score, Ramachandran distribution, rotamer distribution, secondary-structure realism.
Ensemble thermodynamics	RMSF, pairwise distance distributions, contact probability maps, torsion distributions, free-energy surfaces, metastable-state populations.
Kinetics	Implied timescales, transition matrices, Chapman-Kolmogorov tests, mean first passage times, contact autocorrelation, domain-angle autocorrelation, transition path diversity.
Transfer	Unseen trajectory, unseen protein, unseen fold, unseen length, unseen temperature, mutation, ligand state, force-field shift.

A model that passes structural and ensemble tests is an ensemble generator. A model that also passes kinetic tests is a dynamics simulator. A model that generalizes across proteins, folds, temperatures, mutations, and ligands begins to qualify as a world model.

10.2 Anti-noise null controls

Control	Experiment	Expected failure mode if dynamics are learned correctly
Time-shuffled control	Shuffle frames within each trajectory and train the same model.	Structural ensemble may look plausible, but VAMP score, transition matrices, implied timescales, and CK tests should fail.
Random-future control	Train on x_t paired with random x_s from the same protein instead of $t+\tau$.	Equilibrium distribution may be partially recovered, but conditional dynamics and autocorrelations should fail.
Noise-only control	Train on artificial thermal perturbations around one minimized structure.	Model should learn local vibration only and should not hallucinate metastable transitions.
Protein-identity leakage control	Remove or randomize identity features and test unseen proteins/folds.	Performance should not rely on memorized protein identity.
Lag-randomization control	Randomize τ labels.	Displacement scaling, uncertainty scaling, and kinetic observables should degrade.
Temperature-randomization control	Randomize temperature labels.	Temperature-dependent flexibility and rates should degrade.

10.3 Success criteria for meaningful dynamics

- 1 Generated structures are chemically valid.
- 2 Generated ensembles match MD distributions better than simple baselines.
- 3 Slow latent variables correlate with known collective variables and metastable states.
- 4 Transition matrices match MD-derived transition matrices.
- 5 Implied timescales are reasonable across lags.
- 6 Chapman-Kolmogorov tests pass approximately.
- 7 Time-shuffled, random-future, and noise-only controls fail on kinetic metrics.
- 8 Uncertainty increases for unfamiliar folds, ligand states, temperatures, or force fields.
- 9 The model generalizes to proteins and folds excluded from training.
- 10 The model can be used as a proposal mechanism for accelerated sampling with correction.

11. Implementation roadmap

The project should progress from a kinetically meaningful backbone model to side-chain and all-atom extensions. Starting directly with explicit solvent all-atom trajectories would obscure the central question and greatly increase engineering burden.

11.1 18-24 month staged plan

Milestone	Goal	Key output
1. Benchmark and preprocessing	Curate datasets; standardize topology, frames, torsions, contacts, metadata; split by protein/fold.	Dataset object store and train/validation/test protocol.
2. Representation benchmark	Compare raw Cartesian, torsion, residue-frame, contact-map, and hybrid representations.	Representation ablation table with VAMP and validity metrics.
3. Backbone long-stride generator	Train stochastic backbone model conditioned on tau and conditions.	Backbone ensemble generation for unseen proteins.
4. Kinetic latent learning	Add VAMP/Koopman and CK losses.	Improved implied timescales and transition matrices.
5. Side-chain dynamics	Add chi-angle transition and rotamer sampling.	Valid side-chain packing and contact recovery.
6. All-atom refinement	Add all-heavy-atom reconstruction, geometry projection, clash filtering.	Chemically valid all-atom long-stride proposals.
7. Correction and reweighting	Use generated jumps as proposals followed by relaxation/reweighting/MCMC-style acceptance.	More stable equilibrium and reduced rollout drift.
8. World-model transfer	Evaluate unseen folds, mutations, temperatures, ligand states, and lengths.	Evidence for transfer rather than memorization.

11.2 Active-learning loop

- 1 Train the initial model on public MD datasets and curated simulations.
- 2 Generate long-stride trajectories for unseen proteins.
- 3 Detect high-uncertainty, high-disagreement, or low-validity regions.
- 4 Launch targeted MD simulations from those states.
- 5 Add new trajectories to training data with condition labels.
- 6 Retrain or fine-tune the model.
- 7 Repeat until uncertainty and kinetic errors decrease in target regimes.

This turns the model into both a simulation accelerator and an experiment designer.

12. First prototype specification

The first serious prototype should test the core idea without being overwhelmed by all-atom and explicit-solvent complexity. Recommended first version:

Backbone residue-frame graph + soft contact map + torsion features + SE(3)-equivariant encoder + flow-matching transition model + VAMP/Koopman slow-latent head

12.1 Input and output

Component	Specification
Input	Residue frames R_i, t_i ; phi/psi/omega torsions; residue identity; soft contact map; dynamic graph; temperature; stride tau; optional short history.
Output	Future residue frames; future phi/psi/omega torsions; future soft contact map; uncertainty score.
Model	Equivariant residue graph encoder, slow/fast latent split, flow-matching or diffusion transition, geometry-constrained decoder.
Loss	Flow/diffusion transition, torsion, contact-map, VAMP/Koopman, CK consistency, geometry regularization, uncertainty calibration.
Evaluation	Structural validity, RMSF, contact maps, free-energy surfaces, implied timescales, CK tests, null controls, unseen protein/fold transfer.

12.2 Minimal experiments

- 1 Representation ablation: Cartesian vs torsion vs residue-frame vs contact-map vs hybrid.
- 2 Lag ablation: train/test at 10 ps, 100 ps, 1 ns, 10 ns, and 100 ns where available.
- 3 Loss ablation: generative only vs generative + VAMP vs generative + VAMP + CK + equilibrium.
- 4 Null controls: time-shuffled, random-future, lag-randomized, temperature-randomized.
- 5 Transfer split: same protein/new trajectory, new protein/same fold, new fold.
- 6 Rollout test: one-step vs repeated long-stride inference with branching and pruning.
- 7 Correction test: generated proposal alone vs proposal plus geometry/energy/short-MD correction.

13. Risk mitigation

Risk	Why it matters	Mitigation
Deterministic collapse	Exact long-horizon prediction is impossible.	Model conditional distributions and ensembles, not single futures.
Pretty but nonphysical structures	Generative models may produce plausible-looking invalid molecules.	Use geometry constraints, energy filters, and relaxation.
Wrong kinetics	Equilibrium samples do not guarantee correct rates.	Use VAMP/Koopman losses, CK tests, transition matrices, and implied timescales.
Force-field bias	MD data reflects simulation assumptions.	Use multiple datasets/force fields and experimental calibration.
Rare-event under-sampling	Meaningful transitions may be absent in ordinary MD.	Use enhanced sampling, high-temperature data, transition paths, and active learning.
Rollout drift	Repeated generative jumps can accumulate error.	Use branching, pruning, correction, uncertainty, and multi-step training.
No real transfer	Model may memorize training proteins.	Split by family/fold, control identity leakage, and test extrapolation.

Risk	Why it matters	Mitigation
Overfitting to backbone	Backbone can look good while chemistry fails.	Progressively add side chains, all-atom geometry, and packing metrics.

14. Proposed first paper structure

Working title: A Transferable Long-Stride Generative World Model for Protein Molecular Dynamics.

Main claim: A geometry-aware stochastic transition model can generate long-stride protein conformational dynamics for unseen proteins while preserving structural validity, ensemble statistics, and selected kinetic observables.

Figure	Purpose
Figure 1	Concept: femtosecond MD integration versus learned long-stride transition operator.
Figure 2	Representation: residue frames, torsions, graph edges, slow latent variables.
Figure 3	Architecture: encoder, slow/fast latent split, stochastic propagator, decoder, physics heads.
Figure 4	Structural validity and ensemble recovery on unseen proteins.
Figure 5	Kinetic validation: implied timescales, transition matrices, CK test.
Figure 6	Transfer: unseen folds, mutations, temperatures, ligand states.
Figure 7	Acceleration, correction, rollout stability, and failure cases.

14.1 Manuscript logic

- 1 Introduce the timescale gap in MD and the need for transferable long-stride dynamics models.
- 2 Define the transition-kernel formulation and explain why deterministic next-frame prediction is inadequate.
- 3 Present the hierarchical representation and show representation ablations.
- 4 Present the stochastic transition model and coarse-to-fine decoder.
- 5 Show structural validity and ensemble thermodynamics.
- 6 Show kinetic recovery and anti-noise null controls.
- 7 Show transfer to unseen proteins/folds/conditions.
- 8 Discuss limitations, correction mechanisms, and active-learning extension.

15. Appendices

15.1 Representation checklist

- Parse atom/residue topology.
- Build residue frames.
- Compute backbone torsions.
- Compute side-chain torsions.
- Build covalent graph.
- Build dynamic contact graph.
- Compute secondary structure.
- Compute hydrogen bonds and salt bridges.
- Compute slow collective variables.
- Store temperature, force-field, solvent, ligand, mutation, and stride metadata.
- Create train/test splits by protein family and fold.

15.2 Dynamics checklist

- Train backbone-only stochastic model.
- Add stride conditioning.
- Add multi-lag training.
- Add VAMP/Koopman slow-latent loss.
- Add CK consistency.
- Add side-chain torsion decoder.
- Add all-atom reconstruction.
- Add geometry projection.
- Add energy/clash filtering.
- Add uncertainty estimation.
- Add active-learning loop.

15.3 Anti-noise checklist

- Train time-shuffled control.
- Train random-future control.
- Train noise-only control.
- Randomize lag labels.
- Randomize temperature labels.
- Evaluate VAMP score.
- Evaluate implied timescales.
- Evaluate CK consistency.

- Evaluate transition matrices.
- Evaluate multi-step rollout drift.

15.4 Key notation

Symbol	Meaning
x_t	Protein state at time t .
τ	Long stride or lag time.
c	Global condition vector.
h_t	History or memory state.
z_{t_slow}	Latent slow kinetic/conformational state.
z_{t_fast}	Latent fast stochastic/noise state.
K_τ	Transition operator at lag τ .
C_{ij}	Soft contact probability between residues i and j .
D_{ij}	Pairwise distance.
T_i	Residue local frame.

15.5 Selected references verified for this plan

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- Shoghi et al. Scalable Spatio-Temporal SE(3) Diffusion for Long-Horizon Protein Dynamics / STAR-MD. arXiv:2602.02128.
- Mardt, Pasquali, Wu, and Noe. VAMPnets: Deep learning of molecular kinetics. arXiv:1710.06012.
- Garcia Satorras, Hoogeboom, and Welling. E(n) Equivariant Graph Neural Networks. arXiv:2102.09844.
- Jing, Berger, and Jaakkola. AlphaFold Meets Flow Matching for Generating Protein Ensembles. arXiv:2402.04845.
- Jing, Stark, Jaakkola, and Berger. Generative Modeling of Molecular Dynamics Trajectories / MDGen. arXiv:2409.17808.

15.6 Final recommendation

The recommended first model is a multi-resolution, SE(3)-equivariant, stochastic transition model with a kinetic slow latent space. In condensed form:

protein graph + residue frames -> slow latent variables -> flow/diffusion long-stride transition -> geometry-constrained decoder -> physics correction

The central scientific contribution should not be framed as predicting the next MD frame. It should be framed as learning a transferable, physically constrained transition operator that advances protein conformational ensembles over long time strides while preserving thermodynamic and kinetic structure.